
SIGN-AWARE ANTITHETIC COUPLING FOR SIGNED GRADIENT-PARTICLE METHODS

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ABSTRACT

Signed gradient-particle methods recover a field by cumulatively summing positive and negative particle masses. Averaging repeated simulations reduces sampling error, but the benefit of pairing their random increments depends on both the mass signs and the particle positions. We study a coupling that matches two simulations by spatial rank and reflects their Gaussian increments when the matched masses have the same sign. Opposite-sign masses instead receive the same increment. The construction specializes established coupling principles while preserving the probability law of each simulation. We derive an exact reduction in conditional integrated variance for the final diffusion stage, with contributions determined by mass products and particle separations. We then compare the complete nonlinear evolution with reflected pairing, within-sign pairing, control variates, and a randomized quasi-Monte Carlo adaptation. Viscous Burgers and cubic-flux tests show an additional reduction in field variance over within-sign pairing. Separate experiments with plans selected before evaluation show lower execution time at specified expected-error targets. Heat and one-sign controls distinguish the effect of the coupling from nonlinear transport. The benefit depends on the initial profile, observation time, and requested observable, and decreases as the common discretization bias dominates. These results support sign-aware spatial matching for repeated estimation of transient fields by one-dimensional signed gradient-particle solvers.

Keywords gradient random walk · signed particles · antithetic coupling · variance reduction · viscous conservation laws · Monte Carlo

1 Introduction

Gradient-particle methods approximate the spatial derivative of a solution by a signed particle measure and recover the field by cumulative summation. This representation concentrates particles near steep gradients and supports a mesh-free field reconstruction. Gradient random walks have been developed for reaction–diffusion and viscous conservation equations [1–5]. More recently, Bertaglia et al. [6] extended gradient-based Monte Carlo constructions to relaxation approximations of hyperbolic conservation laws, including systems. The present work concerns repeated estimation of a transient field within an existing one-dimensional signed-particle solver.

The computational question is how to couple the randomness in two replicas. Independent averaging reduces sampling variance but requires additional trajectories. Antithetic sampling instead seeks a negative covariance between the outputs of two runs with unchanged marginal laws [7, 8]. In a signed cumulative reconstruction, opposite masses reverse the response to a particle displacement. Reflection of all matched Gaussian increments therefore need not give a favorable field covariance, even when the particles are close in space.

We match particles by spatial rank and reflect their increments when the matched masses have the same sign. When their signs differ, we use synchronous increments. A natural alternative pairs particles within each sign class and reflect

every matched increment. The comparison between these two sign-aware rules is central. It separates the benefit of retaining spatial matching from the benefit already obtained by respecting mass signs.

1.1 Related particle and coupling methods

The coupling principles have established antecedents. Nüsken and Pavliotis [9] formulate variance reduction through couplings with preserved marginals. Their Proposition 54 and Corollaries 57–58 include reflection/synchronization switches based on products of response-related quantities. Section 5.2 describes repeated sorting and pairwise matching within a collection of sampler copies. Their asymptotic variance analysis motivates such constructions, but does not provide a finite-time variance comparison for the signed interacting solver studied here. The choice of spatial rank and mass sign is our specific realization of these ideas for a cumulative field estimator. It requires no auxiliary equation, although computable switching criteria also occur in the cited special cases.

Within the gradient-particle literature, Lécot [10] combines signed cumulative reconstruction, spatial ordering, and quasi-random diffusion inputs for a Burgers construction. A related treatment appears in Lécot and Ogawa [11]. This is the closest representation-specific precedent for structuring the random inputs. Our randomized quasi-Monte Carlo comparison adapts that diffusion strategy to the transport update used here. It is not a reproduction of the complete published algorithm.

Recent QMC particle constructions also couple noise within an interacting ensemble. Ben Rached et al. [12] prove weak convergence for additive-noise McKean–Vlasov dynamics and develop an antithetic multilevel QMC estimator. Their construction and regularity assumptions differ from our signed rank-dependent solver. Together with array-RQMC work for a Fokker–Planck particle model [13], this places the diffusion adaptation below within a broader literature. Its result cannot rank QMC methods as a family.

Bencheikh and Jourdain [14] study bias and antithetic sampling for mean-field particle approximations. Their equation (3.1) couples a $2N$ -particle system with two N -particle systems to form a particle-number correction, an object different from our same-size replica average. Related convergence and bias analyses [15–19] concern particular underlying particle schemes and assumptions. Preservation of our baseline’s law does not extend those theorems to a different reconstruction or transport convention.

Variance reduction can also act through a control variate, conditional expectation, or a modified particle representation [8, 20–22]. We compare the coupling with two known-mean controls and conditional evaluation of the last diffusion stage. These comparisons concern the selected estimators and their implementations, rather than the best attainable performance of each variance-reduction family.

1.2 Contributions and organization

The contribution has three components. First, we specify a sign-aware spatial-rank coupling for signed cumulative field estimation and establish preservation of each replica’s complete discrete law. This retains the baseline expectation and allows its discretization bias to be studied separately from the covariance introduced by pairing.

Second, we derive an exact final-stage variance identity and identify an unconditional equality case when all masses have one sign. The identity is local to the conditioned diffusion stage. A counterexample shows why an unrestricted coordinatewise-monotonicity argument cannot supply the missing nonlinear terminal comparison.

Third, we quantify the additional computational benefit over within-sign pairing, alongside reflected sampling, controls, and a randomized quasi-Monte Carlo adaptation. Fixed-profile experiments, two distinct held-out target comparisons, and nonlinear observables establish the empirical scope.

The intended use is ensemble field estimation after a signed-particle solver has been selected. We do not compare its efficiency with deterministic PDE solvers. The computations are one-dimensional and scalar, and use the mean-transport update defined below. In particular it does not establish an advantage for sampled two-speed transport.

Section 2 defines the update and estimators. Section 3 gives their exact properties. Section 4 presents the comparisons and applicability tests, followed by the discussion and conclusions. Appendices supply proofs, reference and initialization details, and the statistical methodology.

2 The solver, the estimators and the coupling

2.1 Problem class and signed particle representation

We consider the scalar viscous conservation law

$$\partial_t u + \partial_x f(u) = \nu \partial_{xx} u, \quad u(x, 0) = u_0(x), \quad x \in \mathbb{R}, \quad 0 < t \leq T. \quad (2.1)$$

We take $f \in C^1(\mathbb{R})$ and $\nu > 0$, with initial data u_0 of bounded variation whose derivative is integrable and whose limits $u_{\pm} = \lim_{x \rightarrow \pm\infty} u_0(x)$ exist. The experiments use $f(u) = u^2/2$ (viscous Burgers), $f(u) = u^3/3$, and $f \equiv 0$ (heat), the last as a control in which the coupling acts without transport.

For a smooth solution, differentiating (2.1) gives

$$\partial_t w + \partial_x (f'(u)w) = \nu \partial_{xx} w, \quad w = \partial_x u.$$

Thus the gradient is transported with velocity $f'(u)$ and diffused with coefficient ν . The signed-gradient representation used in random-walk methods approximates w by weighted point masses and recovers u by integration (see [1, 4, 10]). We write

$$\begin{aligned} w^N(\cdot, t) &= \sum_{i=1}^N m_i \delta_{X_i(t)}, \\ U^N(x, t) &= u_- + \sum_{i: X_i(t) \leq x} m_i. \end{aligned} \quad (2.2)$$

Here δ_{X_i} is the Dirac measure at particle position X_i , and m_i is its signed mass. We keep the masses fixed and evolve the positions. For smooth profiles with equal far-field values, the positive and negative parts of u'_0 have equal integral. The main two-pulse experiments place $N/2$ particles at midpoint quantiles of each part and assign magnitude

$$|m_i| = \frac{\text{TV}(u_0)}{N}, \quad \text{TV}(u_0) = \int_{\mathbb{R}} |u'_0(x)| dx.$$

Here $\text{TV}(u_0)$ is the total variation of the initial field. Equal counts and equal magnitudes enforce $\sum_i m_i = 0$ for this construction. The profile/time study instead normalizes each sign class using its sampled integral, and the one-sign control has nonzero total mass. These numerical initializations are specified in Appendix C. Zero total mass is required for the whole-line product identity in Lemma 3.4, but not for preservation of a replica's law.

2.2 One step of a single replica

The diffusion stage uses the Gaussian random-walk increment of variance $2\nu h$ [1]. We fix a time step $h > 0$ and set

$$K = T/h \in \mathbb{N}, \quad \sigma = \sqrt{2\nu h}.$$

A replica (i.e., one simulation of the particle system) keeps a state (X, m, v) of positions, masses and carried velocities. Its storage order is the most recent pre-diffusion rank order. After diffusion the positions need not be sorted. Algorithm 1 specifies the update used in all comparisons. We first transport with the carried velocity and sort the particles. We then evaluate the new velocity from the cumulative mass at each rank, including that particle's own mass, and apply diffusion. This is our baseline convention. It is not a reproduction of the shock and rarefaction transport rules in Lécot [10].

Algorithm 1: One step of a single replica (mean-transport gradient random walk)

Input: state (X, m, v) with associated entries, step h , viscosity ν , flux derivative f'

- 1 $X_i \leftarrow X_i + v_i h$ // transport with the carried velocity;
 - 2 $\pi \leftarrow$ a stable sort of X ; $(X, m) \leftarrow (X_\pi, m_\pi)$;
 - 3 $v_i \leftarrow f'(u_- + \sum_{j \leq i} m_j)$; // new carried velocity, inclusive cumulative sum
 - 4 draw $Z \sim \mathcal{N}(0, I_N)$ independent of the past; $X_i \leftarrow X_i + \sigma Z_i$;
-

Three features of Algorithm 1 matter later. The velocity is computed *after* the sort and *before* the diffusion, and is used by the *next* transport, so it is part of the state and must stay associated with its particle until used. At the next sort, we discard and recompute that velocity. Any other reordering must permute the velocity together with the position and mass. Particles at the same position retain their previous relative order. We do not sort again between diffusion and the next transport.

The velocity is evaluated from the rank-inclusive cumulative sum. At tied positions this convention uses the stable particle order. It differs from the value of a field reconstruction that includes all co-located masses at once. The same convention is used in every coupling. This mean-transport baseline uses the reconstructed velocity directly, without sampling a velocity label.

Figure 1 shows the two matchings on one paired state. Within-sign matching restricts each particle to a partner of the same sign, which can increase the separation. Spatial-rank matching pairs the i th particle of one replica with the i th of the other and uses the mass signs to choose the increment orientation.

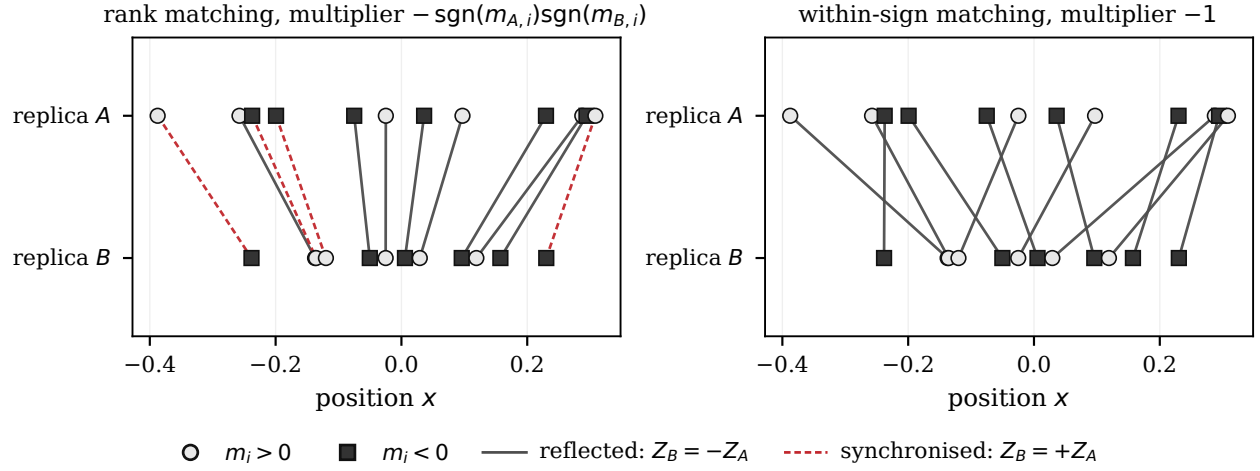


Figure 1: The two matchings applied to the same paired state immediately before the tenth diffusion stage of the interleaved-sign configuration, with $N = 12$. The left panel pairs particles of equal spatial rank and synchronizes the increments of unlike-sign pairs. The right panel restricts partners to the same sign and reflects every pair. The state is identical in both panels, so the difference in separation comes only from the matching rule. Both rules preserve each replica's law (Proposition 3.1).

Implementation requirements. The Gaussian vector must be independent of the complete joint history, and the matching must be a bijection. Reusing one coordinate for two particles of the same replica changes their joint transition law. The multipliers must equal $+1$ or -1 to preserve the increment variance. We exclude zero masses, since inserting $\text{sgn}(0) = 0$ into (2.4) would suppress diffusion for that particle. An implementation retaining zero masses would need an explicit convention, such as setting the multiplier to -1 .

Within-sign matching also requires equal counts in each sign class across the replicas. Our deterministic common initialization and fixed masses ensure this property. These conditions, together with retaining the association between each particle and its carried velocity, specify the full transition whose law is preserved below.

2.4 Estimators, error functional and work

Write $U^{(A)}, U^{(B)}$ for the two reconstructed fields at time T . The unpaired estimator averages B independent simulations,

$$\hat{u}_B^{\text{sing}}(x) = \frac{1}{B} \sum_{b=1}^B U^{(b)}(x). \quad (2.5)$$

For coupled simulations we first average the two fields in each pair,

$$\bar{U}^{(b)}(x) = \frac{1}{2} (U^{(A,b)}(x) + U^{(B,b)}(x)), \quad (2.6)$$

and then average B independent realizations of the pair,

$$\hat{u}_B(x) = \frac{1}{B} \sum_{b=1}^B \bar{U}^{(b)}(x). \quad (2.7)$$

Thus B counts single simulations in (2.5) and pairs in (2.7). A paired estimate requires $2B$ trajectories.

We reconstruct the fields at $L = 400$ equispaced points x_ℓ in $\Omega = [-5, 5]$, with spacing Δx . The discrete approximation of integrated squared error is

$$\mathcal{E}(B) = \mathbb{E} \|\hat{u}_B - u_{\text{ref}}\|_\Omega^2, \quad \|g\|_\Omega^2 = \Delta x \sum_{\ell=1}^L g(x_\ell)^2. \quad (2.8)$$

Here u_{ref} is an independently computed spectral or Cole–Hopf solution, specified in Appendix C. Independence across replicates gives the bias–variance decomposition

$$\mathcal{E}(B) = b^2 + \frac{V}{B}, \quad (2.9)$$

where the squared bias and integrated variance are

$$b^2 = \|\mathbb{E}\bar{U} - u_{\text{ref}}\|_{\Omega}^2, \\ V = \Delta x \sum_{\ell=1}^L \text{Var}(\bar{U}(x_{\ell})).$$

The bias includes finite-particle and time-stepping errors, together with any error in the reference. Only the variance term decreases when more independent pairs are averaged at fixed particle count and time step.

We measure computational work by elapsed execution time,

$$W = BC,$$

where C is the measured cost of one replicate. Each experiment states its timing protocol. The cost of selecting a plan from preliminary simulations is reported separately from execution of that plan.

The discrete norm in (2.8) is used for the numerical comparisons. The theoretical identity in Section 3 instead integrates over the whole real line. For nonlinear observables, applying a functional to the pair mean and averaging its values on the two fields give different estimators. Their expectations and total errors are compared in Section 4.5.

3 Properties of the coupled estimator

Law preservation applies to the full discrete trajectory. The variance identity below instead conditions on a single paired state immediately before diffusion. This distinction is necessary because earlier coupling choices change the state on which later stages act.

3.1 Each replica keeps its law

Proposition 3.1 (marginal law preservation). *Let two replicas be advanced by Algorithm 1 from the same initial state under any admissible coupling (Definition 2.1). Then the law of the complete state trajectory $(X_k^A, m_k^A, v_k^A)_{k=0}^K$ of replica A, and likewise of replica B, is exactly that of an uncoupled single replica. Only the joint law of the two changes.*

This argument uses invariance of the Gaussian distribution, as do the adapted orthogonal transformations in Nüsken and Pavliotis [9, Appendix A]. In our proof (Appendix A), a predictable permutation of an i.i.d. Gaussian vector is i.i.d. Gaussian, and predictable ± 1 multipliers preserve $\mathcal{N}(0, \sigma^2)$ coordinatewise, so each replica’s one-step transition kernel is unchanged given the past.

Remark 3.2 (dependence on the baseline update). The proposition requires the conditional Gaussian transition and replica-local deterministic update specified above. In particular it does *not* need the particle signs to be constant in time, nor the output functional to be linear in the particle measure. The present solver keeps signs fixed. A scheme in which masses were updated would still preserve each replica’s law provided the updates are functions of that replica’s own state and the multipliers remain predictable and of unit modulus.

Corollary 3.3 (shared bias). *Under any admissible coupling, $\mathbb{E}\bar{U} = \mathbb{E}U^{\text{sing}}$ pointwise, provided the displayed expectations exist. Thus the squared bias in (2.9) is identical for REFLECT, SWITCH, and WITHIN under the same baseline initialization and update.*

The conclusion follows by linearity of expectation over the two replicas. More generally it holds for the average of any integrable per-replica functional. It does not hold automatically for a nonlinear function of the pair mean. Preservation of the baseline law retains its own valid expectation and bias results, but it neither establishes continuum convergence nor imports error rates proved for other particle methods.

3.2 An exact identity for the final diffusion stage

We express the integrated covariance of cumulative fields in terms of particle separations. The following product identity supplies that step. Its zero-mass assumption ensures that the fields vanish outside a bounded interval, so their product is integrable over the real line.

Lemma 3.4 (whole-line product identity). *Let $D_A(x) = \sum_i m_i^A \mathbf{1}\{X_i^A \leq x\}$ and $D_B(x) = \sum_j m_j^B \mathbf{1}\{X_j^B \leq x\}$ with $\sum_i m_i^A = \sum_j m_j^B = 0$. Then D_A, D_B are compactly supported and*

$$\int_{\mathbb{R}} D_A(x) D_B(x) dx = -\frac{1}{2} \sum_{i,j} m_i^A m_j^B |X_i^A - X_j^B|. \quad (3.1)$$

Proposition 3.5 (final-stage identity). *Let $g_s(d) = \mathbb{E}|d + sZ|$ with $Z \sim \mathcal{N}(0, 1)$, and condition on the paired state immediately before the last diffusion stage, with a_i, b_i the positions of the rank- i particles of A and B , $d_i = a_i - b_i$, and masses m_i^A, m_i^B satisfying the hypothesis of Lemma 3.4. Write $V_{\bullet} = \int_{\mathbb{R}} \text{Var}(\bar{U}(x) \mid \text{state}) dx$ for the conditional whole-line integrated variance of the pair mean under coupling $\bullet$. Then*

$$V_{\text{REFLECT}} - V_{\text{SWITCH}} = \frac{1}{4} \sum_{i: m_i^A m_i^B < 0} |m_i^A m_i^B| \left[g_{2\sigma}(d_i) - |d_i| \right] \geq 0. \quad (3.2)$$

Only matched pairs of unlike mass contribute. Each term is proportional to the product of the mass magnitudes and to the excess of $g_{2\sigma}(d_i)$ over $|d_i|$, which is largest when the matched separation is small relative to σ and decays as $|d_i|/\sigma$ grows. Non-negativity is Jensen’s inequality. The proof (Appendix A) uses Lemma 3.4 twice and the observation that all off-diagonal terms $i \neq j$ are unaffected by the coupling, because the two increments involved are then independent under the two rank-matched rules being compared.

Comparison with the final-stage calculation. Equation (3.2) does not say that switching helps whenever the matched signs disagree at an earlier stage. The state distributions at later stages are themselves coupling-dependent. We checked the identity against a direct paired Monte Carlo over the final stage at six pre-final states drawn from the two-pulse problem. All six agreed within a nominal 95% interval built from 12 000 draws in 40 batches, and refining the quadrature window by a factor of 2.7 moved the estimate by 0.02%. Averaging the right-hand side over reflected histories estimates the reduction obtained by switching only their last stage. The conditional mean is unchanged, so the law of total variance gives this comparison exactly. Dividing that estimated reduction by the separately measured full-history reduction gives 3.75%, 3.71%, and 1.95% in three archived Gaussian/two-pulse cells. These are descriptive ratios of estimates, not a guaranteed fraction of the full-history gain. The identity therefore explains a local improvement but leaves most of the measured terminal comparison to numerical investigation.

3.3 When the two couplings are identical

Lemma 3.6 (conditional identity). *Start both couplings from the same initial state and drive them with the same sequence of draws $Z^{(k)}$. Then their complete states coincide at every stage through the pre-diffusion state of the first stage at which a matched pair has unlike masses. Their diffusion updates at that stage can differ. In particular, if no such stage occurs before K , the two runs are identical.*

While $\text{sgn}(m_i^A) \text{sgn}(m_i^B) = +1$ for all i , the multiplier (2.4) equals -1 , which is the reflected multiplier. Induction over the stages gives the claim.

Corollary 3.7 (one-sign case). *If every gradient mass has the same sign, the hypothesis of Lemma 3.6 holds for every history, and SWITCH and REFLECT are the same algorithm.*

For monotone initial data, all gradient masses have the same sign. Corollary 3.7 therefore shows that switching makes no change to reflected pairing in this case. Reflected pairing itself can still reduce variance relative to independent runs. We checked equality of positions, masses, and carried velocities for 200 seeds under both heat and Burgers dynamics, with maximum absolute difference exactly 0 in every seed.

Initially separated signs. Finite separation of the positive and negative blocks does not ensure equality of the two couplings. Gaussian increments can reverse their order. For two particles initially separated by $d > 0$, a reflected pair of simulations has exactly one order reversal with probability

$$\Pr(|Z_1 - Z_2| > d/\sigma) > 0.$$

The resulting mismatch of signs at equal ranks can make subsequent switched and reflected updates differ. In the separated configuration at $N = 2048$, the complete states diverged in $58/200 = 0.290$ $[0.228, 0.358]$ of fresh seeds under heat dynamics and $59/200 = 0.295$ $[0.233, 0.363]$ under Burgers. The brackets are exact binomial intervals. Among the runs that diverged, the median first divergence occurred at step 187 of 200.

3.4 No unrestricted monotonicity

Remark 3.8 (coordinatewise monotonicity fails). Coordinatewise monotonicity of a response is a standard sufficient condition for antithetic variance reduction [8]. The corresponding orientation of the terminal field in each particle’s initial position fails for Algorithm 1. With masses $(1/2, -1/2)$, $h = 0.005$ and $\nu = 10^{-10}$, moving the positive particle right from -0.00275 to -0.00225 , while keeping the negative particle at zero, increases the expected field at $x = 0.00125$ after two steps. Appendix B gives the exact zero-noise trace and a Gaussian-tail bound establishing the strict inequality for positive viscosity.

This counterexample excludes the unrestricted monotonicity argument for the specified discrete update. It does not compare the terminal variances of SWITCH and REFLECT, and does not exclude a result under additional restrictions on particle spacing, time step, or viscosity. Its transport-to-diffusion ratio $|m|h/\sigma = 2500$ lies far outside the main experiments.

3.5 A geometric fact about rank matching

To compare the geometry of the two matchings, we measure the mean distance between matched particles at the same paired state. The following standard property of sorted matching fixes the direction of that comparison.

Remark 3.9 (sorted matching is the optimal matching for absolute distance). For two equally sized finite sets of reals $\{a_i\}$ and $\{b_j\}$ listed in increasing order, the identity bijection minimises

$$\sum_i |a_i - b_{\pi(i)}|$$

over all bijections π . This is the monotone rearrangement, which is optimal for convex displacement costs [23]. Rank matching *is* that bijection, and within-sign matching is a bijection constrained to respect the sign classes. Hence

$$\sum_i |a_i - b_i| \leq \sum_i |a_i - b_{\pi_{\text{WITHIN}}(i)}|$$

with probability one, and the ratio of mean matched separations reported in Table 4 is bounded above by 1 by construction.

The separation ratio measures how much the sign constraint increases the mean distance between partners. Proposition 3.5 relates distance to the conditional variance reduction for rank matching. It does not compare rank matching with within-sign matching, whose covariance also depends on the different partners. We therefore use separation to describe the geometry, rather than to infer the cause of a terminal variance difference.

4 Numerical study

4.1 Test configuration and statistical intervals

Unless stated otherwise the test problem is the asymmetric signed two-pulse initial condition

$$u_0(x) = 0.8 e^{-(x+0.7)^2/(2 \cdot 0.5^2)} - 0.5 e^{-(x-0.4)^2/(2 \cdot 0.3^2)}, \quad (4.1)$$

with $f(u) = u^2/2$, $\nu = 0.1$, $T = 1$ and $h = 0.005$ ($K = 200$ steps), evaluated on $\Omega = [-5, 5]$ at $L = 400$ points. It has three sign regions in the numerically resolved u'_0 , so both sign classes are populated and the classes interleave once. Reference solutions are Fourier pseudospectral on $[-16, 16]$ with $M = 2048$ modes, $\frac{2}{3}$ dealiasing and an integrating-factor RK4 step of 10^{-4} . For Burgers this agrees with a stabilised whole-line Cole–Hopf evaluation to 5.6×10^{-14} in the maximum norm, and for the cubic flux it self-converges to 1.6×10^{-13} (Appendix C).

All reported experiments use recorded deterministic seed keys. Independent replicates use distinct streams. The archive specifies the complete key tuple for each experiment, including whether the comparison arms share a stream. Bootstrap resampling uses complete realizations rather than individual reconstruction points [24]. Comparisons with shared streams resample the corresponding realizations together, preserving their dependence. The main field, control and nonlinear-MSE tables use 6000 percentile-bootstrap resamples. The profile/time and structure tables retain their archived 3000 and 2000 resamples, respectively. Scalar station variance ratios use $F(n-1, n-1)$ intervals for independent arms. Held-out errors use t intervals on 32 independent block errors, with Bonferroni correction across the arms of each experiment. Divergence frequencies use exact binomial intervals. The intervals are approximate except where an exact model-based construction is specified. Appendix D gives an empirical scalar calibration and its limits. It does not establish coverage for every field or MSE interval.

Throughout, a ratio below 1 favours the numerator and an interval containing 1 means the comparison is *unresolved at this sample size*. Such an interval does not establish absence of an effect.

4.2 Comparison with alternative estimators

Within-sign pairing already avoids matching opposite masses with reflected increments. We use it as the main comparator to determine whether retaining spatial-rank matching gives a further reduction in variance. Reflected pairing provides the common baseline.

Table 1 reports the integrated variance and mean squared error of the pair mean. Within-sign pairing reduces the variance to 0.172 of the reflected value. Sign-switched rank pairing reduces it to 0.127, or 0.739 of the within-sign value. The interval for this additional reduction excludes 1.

Table 1: Integrated variance and integrated mean squared error over $\Omega = [-5, 5]$ for the pair mean, two-pulse Burgers, $N = 8192$, $h = 0.005$, 48 replicates, with whole-replicate percentile bootstrap intervals. The absolute integrated variances are 4.326×10^{-5} (reflected), 7.433×10^{-6} (within-sign), 5.492×10^{-6} (sign-switched).

comparison	integrated variance ratio	integrated MSE ratio
within-sign / reflected	0.1718 [0.1450, 0.2038]	0.2056 [0.1738, 0.2443]
sign-switched / reflected	0.1269 [0.1073, 0.1511]	0.1684 [0.1425, 0.2007]
sign-switched / within-sign	0.7389 [0.6817, 0.8006]	0.8193 [0.7542, 0.8928]

Figure 2 illustrates cancellation of the two replica errors in one paired realization. Its variance panel, computed over all 48 replicates, shows that the improvement varies substantially with location. These are spatial diagnostics of this field comparison, not independent confirmations or pointwise performance guarantees. Section 4.5 gives separate station comparisons with uncertainty.

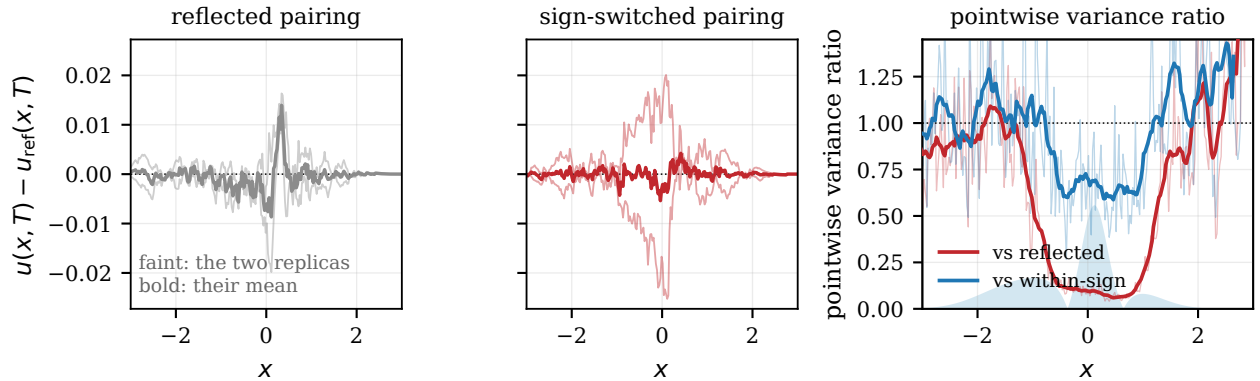


Figure 2: Two-pulse Burgers, $N = 8192$, 48 replicates, from the independently replayed archive. The left and centre panels show the pointwise error of each replica (faint) and of their mean (bold) on one replicate. The right panel shows the ratio of pointwise variance of the sign-switched pair mean to that of the reflected (red) and within-sign (blue) pair means. Thin lines are the raw ratios over 48 replicates and thick lines a 13-point running mean. The shaded profile is $|\partial_x u_{\text{ref}}|$, scaled to the panel.

Cost. A pair costs 1.93 times a single trajectory at $N = 8192$ (0.1950 s against 0.1009 s, serial, from experiment A). In a separate interleaved timing batch, among the three pairings, at $N = 8192$ one replicate costs 0.1941 s (reflected), 0.1995 s (sign-switched) and 0.2117 s (within-sign). The switch costs 1.028 times reflected pairing and 0.943 times within-sign pairing, and at $N = 2048$ the corresponding figures are 1.040 and 0.908. Combining the measured cost with Table 1 gives a variance $\times$ cost ratio of 0.697 against within-sign pairing. The timings describe the implementations on one machine. Within-sign pairing partitions the particles by sign and gathers their partners, but we have not profiled those operations separately and do not attribute its higher elapsed time to a particular operation.

Conditional expectation and control variates. We also apply two standard variance-reduction techniques to the same solver, conditional expectation and control variates [8, Chapter 4]. Table 2 compares them with the three pairings. All arms share seed streams, and the reported intervals retain that pairing.

For conditional evaluation of the last diffusion stage, let (a_i, m_i) be one replica’s pre-final configuration and let Φ denote the standard normal cumulative distribution function. Then

$$\mathbb{E}[\mathbf{1}\{a_i + \sigma Z_i \leq x\} \mid \mathcal{F}_{K-1}] = \Phi\left(\frac{x - a_i}{\sigma}\right).$$

The conditional field is therefore

$$u_- + \sum_i m_i \Phi\left(\frac{x - a_i}{\sigma}\right),$$

and we average this expression over the two replicas. This operation integrates out the final indicator noise, rather than replacing a random displacement by zero.

The heat control and frozen-velocity control use the Brownian displacement W_i accumulated by each particle identity. Their terminal particle positions are, respectively,

$$X_{i,H} = X_i^0 + W_i, \quad X_{i,F} = X_i^0 + v_i^0 T + W_i.$$

Here v_i^0 is the deterministic initial rank velocity, held fixed in the surrogate. The cumulative fields built from these positions have known means

$$\begin{aligned} \mu_H(x) &= u_- + \sum_i m_i \Phi\left(\frac{x - X_i^0}{\sqrt{2\nu T}}\right), \\ \mu_F(x) &= u_- + \sum_i m_i \Phi\left(\frac{x - X_i^0 - v_i^0 T}{\sqrt{2\nu T}}\right). \end{aligned}$$

To obtain these means, observe that the predictable sign and permutation transforms preserve the conditional Gaussian law of every fresh increment. Each identity’s accumulated displacement consequently has distribution $\mathcal{N}(0, 2\nu T)$.

Writing $\bar{H}$ for the average heat-control field of the two replicas gives the control estimator

$$\bar{U}_c(x) = \bar{U}(x) - c(\bar{H}(x) - \mu_H(x)).$$

The frozen-velocity estimator is defined in the same way. We examine $c = 1$ and a fitted scalar coefficient. On a disjoint training sample, the latter is

$$c^* = \frac{\sum_\ell \widehat{\text{Cov}}(\bar{U}(x_\ell), \bar{H}(x_\ell))}{\sum_\ell \widehat{\text{Var}}(\bar{H}(x_\ell))}.$$

This minimizes the sample integrated variance over scalar coefficients. The common factor Δx cancels between numerator and denominator. We hold the fitted coefficient fixed during evaluation and report its training cost separately.

Table 2: Alternatives on the same solver. Integrated variance relative to reflected pairing, with paired whole-replicate bootstrap intervals, at $N = 1600$, $h = 0.005$, 64 seeds. “ $\times c$ ” multiplies the variance ratio by the measured per-replicate cost ratio. One-off coefficient training, where required, is listed separately and is not amortised in $\times c$. c^* denotes a coefficient fitted on a disjoint batch and frozen.

flux	arm	variance / reflected	$\times c$	one-off (s)	variance / within-sign
Burgers	reflected	1.0000	1.000	–	–
	+ conditional final stage	0.8813 [0.852, 0.909]	1.612	–	–
	+ heat control, c^*	0.8440 [0.821, 0.869]	1.170	0.488	–
	+ frozen-velocity control, c^*	0.6724 [0.617, 0.740]	0.923	0.466	–
	within-sign	0.2083 [0.179, 0.245]	0.243	–	1
	sign-switched	0.1649 [0.142, 0.194]	0.174	–	0.7916 [0.762, 0.825]
	+ frozen-velocity control, c^*	0.1642 [0.141, 0.194]	0.237	0.490	0.7886 [0.758, 0.822]
	+ conditional final stage	0.0804 [0.069, 0.095]	0.151	–	0.3861 [0.362, 0.412]
cubic	reflected	1.0000	1.000	–	–
	+ frozen-velocity control, c^*	0.5031 [0.458, 0.552]	0.697	0.448	–
	within-sign	0.2263 [0.196, 0.262]	0.268	–	1
	sign-switched	0.1739 [0.151, 0.201]	0.184	–	0.7688 [0.741, 0.797]
	+ heat control, c^*	0.1684 [0.146, 0.194]	0.248	0.496	0.7442 [0.716, 0.773]
	+ conditional final stage	0.0818 [0.070, 0.096]	0.158	–	0.3614 [0.340, 0.383]

In Table 2, the frozen-velocity control on reflected pairing gives relative variance 0.672 but variance $\times$ cost 0.923 on Burgers, before its 0.47 s of coefficient training. At this configuration the switched coupling has lower variance

and lower variance times cost. Adding the fitted frozen-velocity control to the switched coupling changes its relative variance from 0.1649 to 0.1642 on Burgers. On the cubic flux, adding the heat control changes it from 0.1739 to 0.1684, a resolved ratio of 0.968. This is a statement about these two controls at these configurations and not a claim that no control can help.

Conditional averaging can be combined with the switched coupling. On Burgers it reduces relative variance from 0.1649 to 0.0804 and gives the smallest variance times cost in the table (0.151). Its computational benefit depends on the number of conditional CDF evaluations. When that number increases from 201 to 1601, the measured variance–cost ratio for adding the conditional stage changes from 0.614 to 1.485. Thus lower variance can come with greater work. We report the coupling and conditional evaluation separately to distinguish their contributions.

A randomized quasi-Monte Carlo adaptation of Lécot [10]’s diffusion construction applied to our transport solver is included in the held-out experiment in §4.4.1. It attained the target with 46 trajectories against four.

4.3 Dependence on profile, time, and flux

We next vary the initial profile, observation time, and flux to test the scope of the fixed-setting comparison.

Profiles and times. Three fixed initial profiles have different sign structures. We use a single bump (2 regions), the two-pulse profile (4.1) (3), and an oscillatory profile $0.45 \sin(3x)e^{-x^2/(2 \cdot 1.1^2)}$ (13). Each is observed at $T = 0.25, 1.0$ and 2.0 with $N = 2048$ and 48 replicates. Table 3 and the left panel of Figure 3 give the result. The region counts use the magnitude threshold specified in Appendix C.

Table 3: Integrated variance ratio of sign-switched to within-sign pairing, $N = 2048$, $h = 0.005$, 48 replicates, whole-replicate bootstrap intervals.

profile	sign regions	$T = 0.25$	$T = 1.0$	$T = 2.0$
single bump	2	0.958 [0.886, 1.039]	0.790 [0.727, 0.862]	0.810 [0.735, 0.893]
two-pulse	3	0.895 [0.830, 0.967]	0.785 [0.713, 0.866]	0.793 [0.702, 0.889]
oscillatory	13	0.932 [0.876, 0.990]	0.667 [0.613, 0.725]	0.595 [0.544, 0.655]

The intervals in eight of the nine configurations in Table 3 lie below 1. The exception is the single bump at $T = 0.25$, whose interval [0.886, 1.039] leaves the comparison unresolved. All three profiles show a larger reduction at $T = 1$ and $T = 2$ than at $T = 0.25$. The ratios need not decrease at every observation time. Between $T = 1$ and $T = 2$, they increase from 0.790 to 0.810 for the single bump and from 0.785 to 0.793 for the two-pulse profile. The oscillatory profile has the largest reduction at the two later times.

The profiles also differ in amplitude, shape, and gradient magnitude. Their ordering therefore does not isolate sign-region count as the cause. We examine the matching geometry separately by applying both matching rules to the same paired state.

Table 4: Mean matched-pair separation under rank matching divided by that under within-sign matching, both rules applied to the *same* paired state so the history is held fixed, with $N = 2048$, 24 replicates, ± 2 standard errors. By Remark 3.9 every entry is at most 1 with probability one.

profile	sign regions	$T = 0.25$	$T = 1.0$	$T = 2.0$
single bump	2	0.928 [0.908, 0.948]	0.851 [0.817, 0.885]	0.790 [0.741, 0.840]
two-pulse	3	0.897 [0.879, 0.916]	0.839 [0.803, 0.875]	0.820 [0.762, 0.878]
oscillatory	13	0.847 [0.832, 0.862]	0.693 [0.655, 0.732]	0.676 [0.624, 0.728]

Table 4 and the right panel of Figure 3 show how the within-sign constraint increases matched distances. The separation ratio is at most 1 by Remark 3.9, and its smallest values occur for the oscillatory profile at the later times. The two diagnostics do not order all profiles in the same way. At $T = 2$, separation ratios order them as oscillatory, single bump, and two-pulse, whereas variance ratios order them as oscillatory, two-pulse, and single bump. Thus the geometric comparison supports retaining nearby partners but does not establish the cause of the full-history variance reduction.

Flux. Replacing $f(u) = u^2/2$ by $f(u) = u^3/3$ changes the character of the transport. Its characteristic velocity is

$$f'(u) = u^2 \geq 0,$$

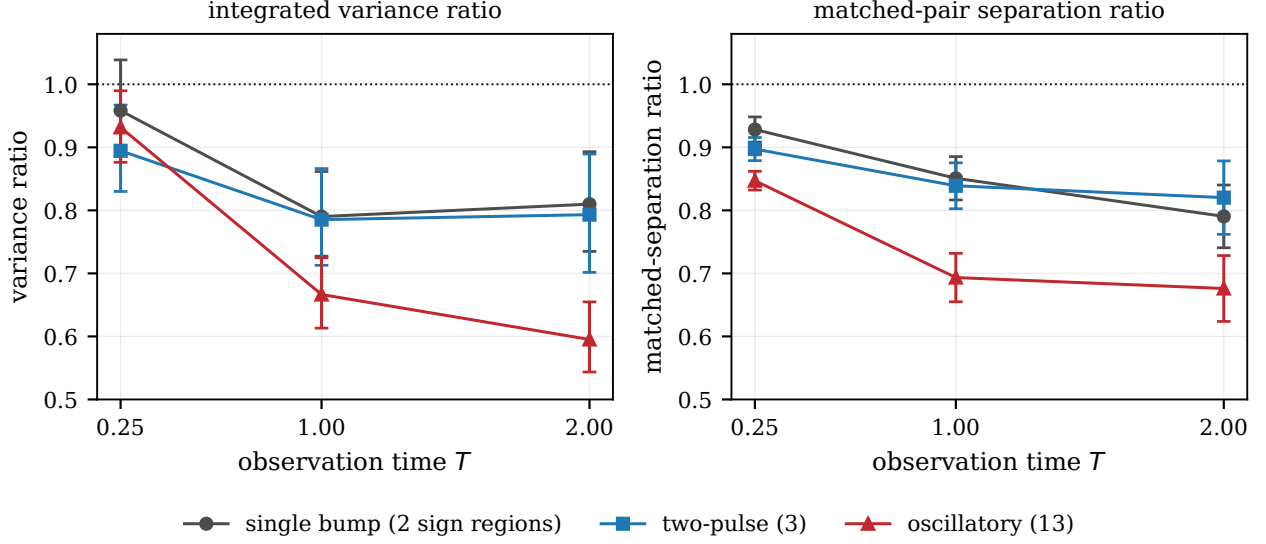


Figure 3: The left panel shows the integrated variance ratio of sign-switched to within-sign pairing across three fixed profiles and three observation times (Table 3). The right panel shows the matched-pair separation ratio at a fixed paired state (Table 4), which cannot exceed 1 by Remark 3.9. It measures the increase in distance from constraining the matching to respect sign classes, not the variance mechanism.

so advection is rightward or stationary, and the velocity depends quadratically on the field. With everything else held fixed at $N = 1600$, $h = 0.005$ and the same 64 seeds, the sign-switched-to-within-sign variance ratio is 0.7916 [0.762, 0.825] for Burgers and 0.7688 [0.741, 0.797] for the cubic flux (Table 2). The advantage does not require bidirectional transport. The comparison tests these two fluxes at the stated configuration.

4.4 Work at selected accuracy targets

To measure execution time at a specified accuracy, we selected plans using preliminary simulations and evaluated them on independent random streams. For each arm, the preliminary sample supplied estimates of squared bias, integrated variance, and cost per replicate. Upper interval endpoints b_{hi}^2 and V_{hi} determined the smallest positive integer B satisfying

$$b_{hi}^2 + \frac{V_{hi}}{B} \leq \frac{\text{tol}}{\kappa}, \quad \kappa = 1.3.$$

The factor κ provides planning headroom. Once chosen, each plan was held fixed and evaluated in 32 independent blocks, where a block is one execution of the complete B -replicate estimator. At a fixed particle count and time step, paired and single arms use the same pilot estimate of squared bias, taken from the switched arm. The randomized quasi-Monte Carlo arm uses its own estimate because its interacting law differs.

The two experiments use different targets, particle counts, and comparators. Experiment A includes an unpaired ensemble and a quasi-Monte Carlo adaptation. Experiment B compares the three pairings, including within-sign matching.

4.4.1 Experiment A with unpaired and quasi-Monte Carlo estimators

Experiment A uses $N = 8192$ and targets expected integrated squared error 7×10^{-6} . The pilot and evaluation seed keys are 1601 and 1702. The target was chosen from the pilot, so it is a pilot-informed target with an independent evaluation batch.

The quasi-Monte Carlo arm is an adaptation of Lécot [10]’s rank-based diffusion to our transport solver, not a reproduction of his method. One `scipy.stats.qmc.Sobol` engine with $d = 2$ and 30 bits is created per replicate and scrambled once. Successive time steps consume successive blocks of N points, as in his equation (45). SciPy’s `scramble=True` is a left-matrix scramble combined with a digital random shift, not fully nested Owen scrambling [25, 26]. The first coordinate assigns a rank by $\lfloor Np_1 \rfloor$ and the second supplies the increment $\sigma\Phi^{-1}(p_2)$, with p_2 clipped

to $[2^{-53}, 1 - 2^{-53}]$ and the clip count recorded. N must be a power of two. We do not claim this adaptation preserves the interacting discrete law. Its total error is measured independently, including its own bias. The rank-bijection assertion is checked outside the timed path.

Table 5: Held-out experiment A. Frozen plans, unused evaluation key, 32 blocks, $N = 8192$, target 7×10^{-6} . “Trajectories” counts solver runs, which is $2B$ for a paired arm and B otherwise. The upper bound is a Bonferroni-corrected 95% t bound across the four arms ($t_{31, 1-0.05/8} = 2.652$). Every arm’s bound lies below the target.

arm	B	trajectories	attained error	simultaneous upper bound	measured work (s)
unpaired ensemble	65	65	5.347×10^{-6}	6.743×10^{-6}	6.658
reflected pairing	17	34	5.031×10^{-6}	6.362×10^{-6}	3.340
RQMC adaptation	46	46	4.024×10^{-6}	4.873×10^{-6}	6.848
sign-switched pairing	2	4	5.100×10^{-6}	5.611×10^{-6}	0.407

4.4.2 Experiment B with within-sign pairing

Experiment B uses $N = 2048$ and target 2.5×10^{-5} , with pilot key 2310 and evaluation key 2320. The evaluation uses 32 blocks, and the separately reported tuning cost is 4.05 s. This experiment uses the standard two-pulse configuration rather than the oscillatory profile that favours the construction.

Table 6: Held-out experiment B. Frozen plans, unused evaluation key, 32 blocks, $N = 2048$, target 2.5×10^{-5} , Bonferroni bound across three arms ($t_{31, 1-0.05/6} = 2.531$). Work is end-to-end with interleaved arm order.

arm	B	trajectories	attained error	simultaneous upper bound	measured work (s)
reflected pairing	18	36	1.404×10^{-5}	1.830×10^{-5}	0.615
within-sign pairing	3	6	1.769×10^{-5}	1.936×10^{-5}	0.119
sign-switched pairing	2	4	1.976×10^{-5}	2.203×10^{-5}	0.0715

All three upper bounds in Table 6 lie below the target. The measured work ratio is 0.599 against within-sign pairing and 0.116 against reflected pairing. These are comparisons of the selected plans at the same target, rather than at equal attained error. Reflected pairing has the smallest estimated error, while the switched arm is closest to the target.

Integer allocation matters at these small replicate counts. The switched plan uses two pairs and the within-sign plan uses three. Combining this count ratio with the separately measured cost ratio gives

$$\frac{2}{3} \times 0.908 = 0.605,$$

close to the measured work ratio 0.599. The result shows that the selected switched plan attains the target with one fewer pair. It does not establish the minimum work attainable by either method.

Both experiments are reported in Figure 4. For experiment B, the original evaluation collected errors with sequential arm timing. Reported work comes from a separate interleaved timing of the same frozen plans. Reported errors remain those of the original held-out batch.

4.5 Observable dependence and the bias-dominated regime

4.5.1 Variation across reconstruction points

At $N = 8192$ with 48 replicates on a separate seed key, the variance ratio of sign-switched to reflected pairing at four reconstruction points is

x	variance ratio and interval	location
-2.0	0.636 [0.357, 1.135]	left flank
-0.7	0.143 [0.080, 0.255]	left pulse
+0.1	0.093 [0.052, 0.166]	central gradient region
+1.5	0.735 [0.412, 1.311]	right flank.

These are $F(47, 47)$ intervals. The two flank comparisons are unresolved. The integrated reduction therefore does not describe the benefit at every reconstruction point.

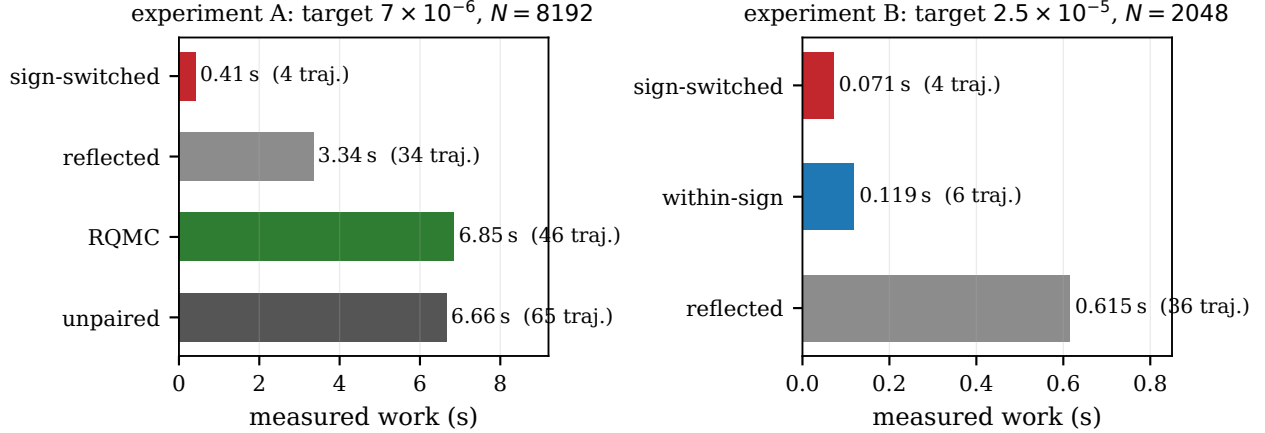


Figure 4: Measured work to attain a common expected-error target in the two held-out experiments, which have different targets, different N and different arms and are not to be pooled. The left panel reports experiment A (Table 5), and the right reports experiment B (Table 6).

4.5.2 Nonlinear functionals of the reconstructed field

Let Q be a nonlinear functional of the field. We distinguish

$$Q(\bar{U}) \quad \text{and} \quad \frac{1}{2}(Q(U^{(A)}) + Q(U^{(B)})).$$

The first applies Q after averaging the fields, whereas the second averages the two values of Q . Their expectations can differ. By Proposition 3.1 the second has exactly the expectation of Q applied to a single uncoupled run, so its bias is the single-run bias under every admissible coupling. The first estimator has no such guarantee. We therefore report the *empirical* mean squared error against the reference,

$$\widehat{\text{MSE}} = \frac{1}{n} \sum_{r=1}^n (Q_r - Q_{\text{ref}})^2 = \widehat{\text{bias}}^2 + \frac{n-1}{n} s^2, \quad (4.2)$$

where $\widehat{\text{bias}} = n^{-1} \sum_r Q_r - Q_{\text{ref}}$ and s^2 is the sample variance with divisor $n-1$. This differs from the plug-in $\widehat{\text{bias}}^2 + s^2$, which exceeds (4.2) by s^2/n (here 0.7%–2.0%). Table 7 uses (4.2) throughout. We evaluate two functionals. The maximum gradient uses central differences at the reconstruction points and depends on their spacing. The level location is the first downward crossing of $u = 0.25$ to the right of the global maximum, found by linear interpolation. We record a realization without such a crossing as censored. None occurred in the reported configurations.

Table 7: Nonlinear observables at $N = 8192$, 48 replicates, against the spectral reference, from the independently replayed archive. Ratios are of the empirical mean squared error (4.2), with whole-replicate bootstrap intervals. The last column is the squared-bias share of the sign-switched arm’s empirical MSE. An interval containing 1 is unresolved.

functional	estimator	switched / reflected	switched / within-sign	bias ² share
$\max \partial_x u $	Q of the mean	0.416 [0.244, 0.702]	0.602 [0.359, 0.984]	0.56
	mean of Q	0.832 [0.603, 1.139]	1.058 [0.803, 1.394]	0.87
level $x(u = 0.25)$	Q of the mean	0.129 [0.070, 0.247]	0.855 [0.497, 1.419]	0.19
	mean of Q	0.090 [0.053, 0.158]	0.847 [0.497, 1.486]	0.30

Table 7 shows that the benefit depends on both the functional and the order of averaging. For the level crossing the switched coupling reduces total error by roughly an order of magnitude against reflected pairing under either estimator. For the peak gradient, using the mean of Q gives ratios 0.832 [0.603, 1.139] against reflected pairing and 1.058 [0.803, 1.394] against within-sign pairing. Both comparisons are unresolved. The plug-in squared-bias share is 87% for that estimator. It is itself estimated, and Corollary 3.3 establishes sharing of the population bias across the couplings, not exact equality of finite-sample estimates. A variance reduction can only reduce the sampling component of this error. The corresponding variance ratios are 0.410 (Q of the mean) and 0.686 (mean of Q) against reflected pairing, giving factors of 2.4 and 1.5.

4.5.3 The bias-dominated regime

The same arithmetic applies to the field itself. Since the couplings share b^2 , (2.9) gives

$$\frac{\mathcal{E}_{\text{SWITCH}}(B)}{\mathcal{E}_{\text{REFLECT}}(B)} = \frac{b^2 + V_{\text{SWITCH}}/B}{b^2 + V_{\text{REFLECT}}/B} \rightarrow 1 \quad \text{as } B \rightarrow \infty, \quad (4.3)$$

when $b^2 > 0$. If $b^2 = 0$, the ratio instead remains $V_{\text{SWITCH}}/V_{\text{REFLECT}}$. Thus the crossover concerns nonzero baseline bias at fixed N and h . The crossover occurs at

$$B^* = V/b^2,$$

where the sampling and squared-bias contributions are equal. At three archived cells of the two-pulse problem B^* for the switched arm is 10.3 ($N = 6400$, $h = 0.0025$), 4.5 ($N = 6400$, $h = 0.005$) and 35.3 ($N = 1600$, $h = 0.0025$). Figure 5 uses equal continuous work instead of equal pair counts:

$$\frac{b^2 + V_{\text{SWITCH}}C_{\text{SWITCH}}/W}{b^2 + V_{\text{REFLECT}}C_{\text{REFLECT}}/W}. \quad (4.4)$$

This is a plug-in model, excluding integer allocation and timing uncertainty. The crossover marks equal sampling and bias contributions in the switched arm. It is not a cutoff for improvement.

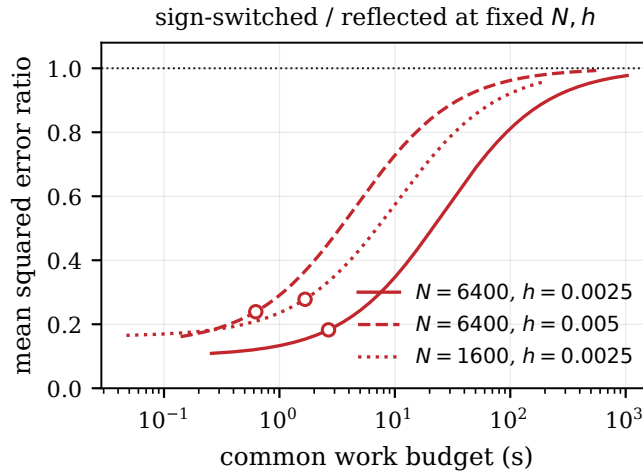


Figure 5: Model mean squared error ratio (4.4) at equal work, at three archived (N, h) cells of the two-pulse problem with 64 paired seeds. The shared squared bias b^2 is the same estimate in numerator and denominator, as Corollary 3.3 requires. Circles mark $W = C_{\text{SWITCH}}V_{\text{SWITCH}}/b^2$. Three configurations do not establish a convergence rate or independently resolve the effects of N and h .

Figure 5 shows the gradual loss of benefit as the common bias becomes a larger fraction of the total error. Coupling reduces the sampling component. Reducing the remaining baseline bias requires changing the particle representation or time discretization.

4.6 Signed-structure controls

Table 8 compares four prescribed arrangements of signed masses under heat and Burgers evolution. Heat removes nonlinear transport, allowing us to examine the coupling under diffusion alone.

Alternating masses. The interleaved configuration uses alternating masses $\pm 1/N$ on a fixed support, giving a reconstructed field whose amplitude is $1/N$. It is 2.5×10^{-3} at $N = 400$ and 1.2×10^{-4} at $N = 8192$, while the total absolute mass stays at 1. Increasing N therefore changes the represented field, as well as the particle count, and its ratios must not be read as convergence under refinement. The merging and separated configurations hold the amplitude at 0.5 and are refinements of one fixed profile.

Separated blocks. The unresolved rows are consistent with the divergence analysis in Section 3. The two couplings agree on about 71% of seeds and, when they do diverge, do so at a median step of 187 out of 200, leaving little history over which a difference can accumulate. This is a statement about this configuration at this N , h , ν and horizon.

Table 8: Integrated variance ratio of sign-switched to reflected pairing by signed structure, $N = 2048$, $h = 0.005$, $T = 1$, 48 replicates, whole-replicate bootstrap intervals. The two arms of each cell were run on *different* seed streams, so these are ratios of independent estimates. The one-sign row is therefore an estimate of a quantity that Corollary 3.7 makes exactly 1, and its departure from 1 is sampling noise. Exact identity is established separately, by a matched-seed full-state comparison.

structure	dynamics	variance ratio	reading
interleaved (alternating signs)	heat	0.049 [0.040, 0.061]	variance reduced
	Burgers	0.063 [0.052, 0.077]	variance reduced
merging blocks	heat	0.476 [0.421, 0.545]	variance reduced
	Burgers	0.481 [0.426, 0.544]	variance reduced
separated blocks	heat	0.994 [0.917, 1.076]	unresolved
	Burgers	1.013 [0.930, 1.101]	unresolved
one sign	heat	0.987 [0.902, 1.078]	exactly 1 by Cor. 3.7
	Burgers	0.967 [0.822, 1.174]	exactly 1 by Cor. 3.7

One sign. The two couplings are the same algorithm (Corollary 3.7). The row checks the sampling variability of the estimated variance ratio. A separate comparison using matched seeds gives zero differences in positions, masses, and carried velocities over 200 seeds for each equation.

5 Discussion and conclusions

Spatial-rank matching with a sign-dependent increment rule reduces the sampling error of signed cumulative field estimates while preserving each replica’s discrete law. The switching principle and adaptive matching have precedents in coupled sampling [9], and spatial sorting with structured diffusion inputs is established for gradient particles [10]. Our contribution specifies their use for paired transient field estimation, derives its final-stage variance identity, and measures the additional benefit over within-sign pairing.

Within-sign pairing accounts for most of the reduction over reflected rank pairing. Retaining spatial matching gives a further reduction in the principal two-pulse experiment, with integrated variance 0.739 times the within-sign value and integrated mean squared error 0.819 times as large. In a separate comparison at a specified expected-error target, the selected switched plan uses four trajectories against six for within-sign pairing and takes approximately 0.60 of its execution time. These plans attain different errors below the target, so the result describes their measured work rather than an optimized equal-error comparison.

The exact identity relates the conditional gain to mass magnitudes, signs, and particle separations at the final diffusion stage. Earlier coupling choices change the distribution of later states, and the full-history comparison remains numerical. Applying both matching rules at a common state shows the additional separation imposed by the within-sign constraint. That geometric property does not establish the cause of the terminal variance reduction, since covariance also depends on which particles are paired.

The profile, time, and cubic-flux comparisons describe a benefit beyond one initial condition. The one-sign identity and unresolved comparisons also identify limits. Coupling is most useful when sampling contributes materially to the requested field error. It cannot remove the common baseline bias, and nonlinear functionals require their own total-error comparisons. The method is therefore an option for ensemble estimation after a signed gradient-particle solver has been selected.

The computations use scalar equations in one dimension, constant particle masses, and mean transport. Sufficient conditions for a terminal variance comparison under nonlinear evolution remain open. Extending the matching construction to systems or several dimensions, and evaluating its cost in parallel implementations, would require separate analysis and experiments.

Data and code availability

The manuscript source, figure-generation scripts, numerical evidence, and reproduction instructions are distributed with the accompanying repository at <https://github.com/stephen122204/rb-gbmc-viscous-burgers>. The evidence map records source files and statistical conventions for each reported result. Historical exploratory outputs are distinguished from the curated evidence used by this manuscript.

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A Proofs

A.1 Proposition 3.1

Let $\Xi_k^A = (X_k^A, m_k^A, v_k^A)$ be the complete state entering stage k , with $k = 0, \dots, K-1$. Let $\mathcal{F}_k$ contain the joint history through the sort and velocity update of that stage, as in Definition 2.1. The deterministic part of Algorithm 1 acts on Ξ_k^A alone. We compare the conditional increment laws before constructing Ξ_{k+1}^A .

Replica A receives $\sigma Z^{(k)}$, with $Z^{(k)} \sim \mathcal{N}(0, I_N)$ independent of $\mathcal{F}_k$. Replica B receives

$$\Delta X_j^B = \sigma \varepsilon_{k,j} Z_{\pi_k(j)}^{(k)}.$$

Given $\mathcal{F}_k$, the matching and sign vector are fixed. A permutation of independent standard Gaussian coordinates preserves their joint law, and multiplication by fixed signs preserves both their distributions and independence. Therefore

$$\mathcal{L}(\Delta X^A \mid \mathcal{F}_k) = \mathcal{L}(\Delta X^B \mid \mathcal{F}_k) = \mathcal{N}(0, \sigma^2 I_N).$$

Each replica has the same conditional transition as the uncoupled solver. Induction over the stages proves equality of its complete trajectory law with the baseline law. The joint law of the two replicas changes because their increments are dependent. $\square$

Independence of the fresh vector and predictability of the rule justify conditioning before the draw. Bijectivity ensures independence within each replica, and unit-modulus multipliers preserve the variance. The particular rule (2.4) also needs nonzero masses. Retaining each particle’s carried velocity in the state makes the deterministic update well defined.

A.2 Lemma 3.4

Fix $c \in \mathbb{R}$ and define the compactly supported difference of indicators

$$h_a(x) = \mathbf{1}\{x \geq a\} - \mathbf{1}\{x \geq c\}.$$

Integrating their product over the overlapping intervals gives

$$\int_{\mathbb{R}} h_a(x) h_b(x) dx = \frac{1}{2} (|a - c| + |b - c| - |a - b|).$$

Since each collection has zero total mass, subtracting the common indicator at c leaves either field unchanged. Thus

$$D_A = \sum_i m_i^A h_{X_i^A}, \quad D_B = \sum_j m_j^B h_{X_j^B}.$$

Expanding the product and applying the preceding identity gives

$$\int_{\mathbb{R}} D_A D_B dx = \frac{1}{2} \sum_{i,j} m_i^A m_j^B (|X_i^A - c| + |X_j^B - c| - |X_i^A - X_j^B|).$$

The first group vanishes because $\sum_j m_j^B = 0$, and the second vanishes because $\sum_i m_i^A = 0$. The remaining term is (3.1). $\square$

A.3 Proposition 3.5

Condition on the pre-final state. Write $D_\bullet$ as in Lemma 3.4 for the terminal configurations, so that $\bar{U} = u_- + \frac{1}{2}(D_A + D_B)$ and

$$V_\bullet = \int \text{Var } \bar{U} = \frac{1}{4} \left(\int \text{Var } D_A + \int \text{Var } D_B + 2 \int \text{Cov}(D_A, D_B) \right).$$

By Proposition 3.1 the first two integrals do not depend on the coupling. For the covariance, apply Lemma 3.4 to the two fields with their coupled noise. Apply it again with independent noise between replicas to obtain the product of their means. Independent Gaussian displacements have difference variance $2\sigma^2$, so subtraction gives

$$\int \text{Cov}(D_A, D_B) = -\frac{1}{2} \sum_{i,j} m_i^A m_j^B \left[\mathbb{E} |X_i^A - X_j^B| - g_{\sigma\sqrt{2}}(a_i - b_j) \right].$$

For $i \neq j$ the increments Z_i and $\varepsilon_j Z_j$ are independent with variance σ^2 each under REFLECT and SWITCH, so $X_i^A - X_j^B = (a_i - b_j) + \sigma\sqrt{2}\mathcal{N}(0, 1)$ in law and the bracket vanishes. Only the diagonal survives. There, under REFLECT, $X_i^A - X_i^B = d_i + 2\sigma Z_i$ and the first term is $g_{2\sigma}(d_i)$. Under SWITCH it is $g_{2\sigma}(d_i)$ when $m_i^A m_i^B > 0$ and, when $m_i^A m_i^B < 0$, the two increments cancel exactly and the first term is $|d_i|$. Subtracting,

$$\begin{aligned} \int \text{COV}_{\text{REFLECT}} - \int \text{COV}_{\text{SWITCH}} &= -\frac{1}{2} \sum_{m_i^A m_i^B < 0} m_i^A m_i^B [g_{2\sigma}(d_i) - |d_i|] \\ &= \frac{1}{2} \sum_{m_i^A m_i^B < 0} |m_i^A m_i^B| [g_{2\sigma}(d_i) - |d_i|]. \end{aligned}$$

and multiplying the covariance difference by $1/2$ gives (3.2). Finally, Jensen's inequality gives

$$g_s(d) = \mathbb{E}|d + sZ| \geq |\mathbb{E}(d + sZ)| = |d|,$$

which establishes non-negativity. $\square$

B A two-particle monotonicity counterexample

Take $u_- = 0$, $m = (1/2, -1/2)$, $h = 0.005$, $\nu = 10^{-10}$, and two steps. The initial positions are $(p, 0)$, with $p_L = -0.00275$ or $p_R = -0.00225$. Both initial carried velocities are $(1/2, 0)$. Without diffusion, the first transport puts the positive particle at -0.00025 in state L and $+0.00025$ in state R . Re-sorting therefore leaves the velocities $(1/2, 0)$ in L but assigns $(-1/2, 0)$ to the negative and positive particles, respectively, in R . After the second transport, the particle positions in identity order are

$$L : (0.00225, 0), \quad R : (0.00025, -0.0025).$$

At the probe $x_* = 0.00125$, the reconstructed fields are $-1/2$ and 0 .

Now restore the Gaussian displacements with $\sigma = 10^{-6}$. On the event that each of the four displacements of a trajectory has modulus at most $\delta = 10^{-4}$, neither its second transport ordering nor its terminal probe classification changes. The relevant ordering gap is $0.00025 > 2\delta$ and the smallest probe margin is $0.001 > 2\delta$. Hence the same field values hold on this event. For either initial state the exceptional probability is at most $\epsilon = 8\Phi(-100) < \exp(-5003)$. Since every field value is in $[-1/2, 1/2]$,

$$\mathbb{E}U_R(x_*, 2h) - \mathbb{E}U_L(x_*, 2h) \geq \frac{1}{2} - \frac{3}{2}\epsilon > 0.$$

The initial movement of a positive mass to the right thus increases the expected terminal field. This proves the claim in Remark 3.8 without inferring an expectation from a finite set of simulation seeds.

C Test problems, initialisation and reference solutions

Initialisation. For smooth profiles with equal far-field values, let

$$w_+(x) = \max(u'_0(x), 0), \quad w_-(x) = \max(-u'_0(x), 0).$$

Both have integral $\text{TV}(u_0)/2$. Their normalized cumulative distributions are

$$F_\pm(x) = \frac{2}{\text{TV}(u_0)} \int_{-\infty}^x w_\pm(y) dy.$$

For each sign class, we place $N/2$ particles at midpoint quantiles,

$$r_k = \frac{k - \frac{1}{2}}{N/2}, \quad X_{k,\pm}^0 = F_{\pm}^{-1}(r_k), \quad k = 1, \dots, N/2,$$

where $F_{\pm}^{-1}$ denotes the generalized inverse. The assigned masses are $\pm \text{TV}(u_0)/N$. This deterministic placement makes the initial state identical across replicas and comparison arms. For the main two-pulse experiments, the code approximates the cumulative integrals by the trapezoid rule on 400 001 points in $[-12, 12]$ and inverts them by linear interpolation. It assigns the same absolute mass to both sign classes.

The profile/time study uses a separate numerical initialization on 200 001 points in $[-6, 6]$. It differentiates the sampled initial profile, forms cumulative rectangle sums of its positive and negative parts, and inverts each cumulative sum by linear interpolation. If M_+ and M_- are those two numerical integrals, the assigned masses are

$$m_{k,+} = \frac{M_+}{N/2}, \quad m_{k,-} = -\frac{M_-}{N/2}.$$

The reconstruction constant is $u_0(-6)$. This construction has equal counts per sign class but does not enforce equality of M_+ and M_- . For the oscillatory profile, its total signed mass is approximately -2.34×10^{-7} . The profile/time comparisons are numerical comparisons of these discrete initial states, and do not invoke the zero-mass identity for the complete evolution.

Profiles. The two-pulse profile is (4.1). The other smooth profiles are

$$\begin{aligned} u_{0,\text{bump}}(x) &= 0.8 \exp\left(-\frac{(x+0.2)^2}{2 \cdot 0.45^2}\right), \\ u_{0,\text{osc}}(x) &= 0.45 \sin(3x) \exp\left(-\frac{x^2}{2 \cdot 1.1^2}\right). \end{aligned}$$

The reported sign-region counts use a 40 001-point evaluation of u'_0 and a relative magnitude threshold of 10^{-6} . This gives 2, 3, and 13 regions for the bump, two-pulse, and oscillatory profiles. The counts refer to gradients above the numerical threshold, rather than to every zero of the analytic derivative on the real line.

The prescribed signed-particle configurations in Section 4.6 use N particles. The interleaved case alternates masses $\pm 1/N$ on $[-0.25, 0.25]$. The separated case places positive masses on $[-2.0, -1.4]$ and negative masses on $[1.4, 2.0]$. The merging case uses the intervals $[-0.9, -0.5]$ and $[0.5, 0.9]$. In the latter two cases there are $N/2$ particles of each sign, with magnitude $1/N$. The one-sign control instead assigns mass $-2/N$ to every particle on $[-1, 1]$.

References. For $f(u) = u^2/2$ and $f(u) = u^3/3$ the reference is a Fourier pseudospectral solution on $[-16, 16]$ with $M = 2048$ modes, $\frac{2}{3}$ -rule dealiasing and integrating-factor RK4 at $dt = 10^{-4}$. For Burgers, the Cole–Hopf transformation supplies an independent whole-line reference [27, 28]. With

$$F(y) = \int_0^y u_0(z) dz,$$

the transformed heat solution and the recovered field are

$$\begin{aligned} \phi(x, t) &= \frac{1}{\sqrt{4\pi\nu t}} \int_{\mathbb{R}} \exp\left(-\frac{(x-y)^2}{4\nu t} - \frac{F(y)}{2\nu}\right) dy, \\ u(x, t) &= -2\nu \frac{\partial_x \phi(x, t)}{\phi(x, t)}. \end{aligned}$$

The potential F is analytic for the Gaussian pulses. A stabilized evaluation agrees with the spectral solution to 5.57×10^{-14} in maximum norm at the reconstruction points. This is a numerical consistency check, not a certified quadrature error bound. For the cubic flux, halving dt and doubling M changes the solution by 1.6×10^{-13} . These reference calculations assess field error. They are not timed competitors and do not enter the fit of a control coefficient.

D Uncertainty methodology and its calibration

Sample quantities and bootstrap intervals. For n independent pair fields, let $\langle \bar{U} \rangle$ denote their sample mean. The estimates corresponding to (2.9) are

$$\hat{V} = \frac{1}{n-1} \sum_{r=1}^n \|\bar{U}^{(r)} - \langle \bar{U} \rangle\|_{\Omega}^2,$$

$$\hat{b}^2 = \|\langle \bar{U} \rangle - u_{\text{ref}}\|_{\Omega}^2 - \frac{\hat{V}}{n}.$$

The subtraction removes the expected sampling contribution to the squared error of the estimated mean. The resulting estimate can be negative even though the population squared bias is nonnegative. We retain its sign in the planning analysis rather than replacing it by zero.

To form a bootstrap interval [24], we draw n realization indices with replacement and recompute the complete statistic, including both parts of any variance ratio. A paired comparison uses the same indices for its arms. Independent arms are resampled independently. The reported percentile interval consists of the 2.5th and 97.5th percentiles of the resampled statistic, with resample counts given in Section 4.1.

Intervals for held-out error. Let e_r be the integrated squared error of the complete estimator in block r , for $r = 1, \dots, 32$. We report the mean $\bar{e}$ and use the upper end of a Bonferroni-adjusted two-sided t interval,

$$U = \bar{e} + t_{31, 1-\alpha/(2J)} \frac{s_e}{\sqrt{32}}, \quad \alpha = 0.05,$$

where s_e is the sample standard deviation of the block errors and J is the number of arms in that experiment. Comparing U with the target assesses expected error under this interval approximation. It does not give the probability that an individual execution falls below the target.

Scalar variance-ratio calibration. The two interval procedures used for scalar observables are a whole-replicate percentile bootstrap and $F(n-1, n-1)$. The second assumes independent normal replicates, which we do not assert, so we calibrated both.

Per-replicate values of four observables were taken from the archived $N = 8192$ fields, *separately* for the reflected and sign-switched arms, and each sample was centred and scaled by its own standard deviation to give two empirical shapes. A synthetic numerator arm resamples the sign-switched shape, scaled to impose a known variance ratio r . The denominator arm resamples the reflected shape. Nominal-95% intervals are then formed by each procedure and coverage counted over $M = 1500$ synthetic datasets, with $n = 48$ per arm and 400 bootstrap resamples per interval. Table 9 reports the two-shape design together with the common-shape design of an earlier calibration, which used one standardised sample for both arms and therefore does not retain the observed difference between the two distributions.

Table 9: Coverage of nominal-95% variance-ratio intervals, $M = 1500$ synthetic datasets per cell, binomial standard error ≤ 0.007 . “Two-shape” resamples the sign-switched and reflected empirical shapes separately. “common-shape” reproduces the earlier design in which both arms came from one standardised sample.

observable	two-shape, $r = 0.15$		two-shape, $r = 0.75$		common-shape, $r = 0.75$	
	bootstrap	F	bootstrap	F	bootstrap	F
station $x = +0.1$	0.921	0.958	0.933	0.967	0.945	0.977
station $x = -2.0$	0.950	0.982	0.946	0.973	0.942	0.977
$\max \partial_x u $	0.941	0.965	0.931	0.960	0.947	0.954
level $x(u = 0.25)$	0.933	0.959	0.928	0.949	0.935	0.963

Under the two-shape design, estimated F coverage is 0.949–0.982 and percentile-bootstrap coverage is 0.921–0.950, with simulation uncertainty as stated in the table. We therefore report F intervals for station and level-crossing variance ratios and bootstrap intervals for integrated-variance and mean-squared-error ratios. The calibration concerns scalar variance ratios. It does not establish coverage for integrated-field or MSE intervals. Those bootstrap intervals are approximate, particularly at these sample sizes. This describes performance under *these* empirical distributions at *this* configuration. It is not a guarantee for other problems, and a non-significant normality test records failure to reject, not normality.

E Evidence provenance

Table 10 maps every table and figure to the archived output that produced it. Paths are relative to the study repository’s output/ directory unless stated otherwise. Each directory name carries a date suffix (`_2026_09_09` or `_2026_09_10`) which is omitted here for width. The file `analysis/numbers.json` contains each quantity together with the archive it was read from.

Table 10: Provenance of reported results.

item	what it answers	archive
Table 1, Fig. 2	principal field comparison	independent replay of <code>round22_corrections</code> ; <code>replayed_fields.npz</code> , <code>replay_audit.json</code>
cost figures, §4.2	what the coupling costs	<code>round22_corrections/costs.json</code> ; <code>round17_compare/compare.json</code> (pair versus single)
Table 2	against controls and a second flux	<code>round09_costs_control/fields.npz</code> , <code>.../rows.csv</code>
Table 3, Fig. 3 (left)	profiles and times	<code>round23_mechanism/mechanism.json</code>
Table 4, Fig. 3 (right)	matched-pair geometry	<code>round24_separation/separation.json</code> (<code>matched_state</code> rows; the <code>own_history</code> rows are the association-only variant, one of nine cells containing 1)
Table 5, Fig. 4 (left)	held-out work, experiment A	<code>round17_compare/compare.json</code>
Table 6, Fig. 4 (right)	held-out work, experiment B	<code>round23_mechanism/worktarget.json</code> ; interleaved timings from <code>round24_separation/timing_check.json</code>
station ratios, §4.5	where the benefit sits	<code>round18_observable/observable.json</code> , <code>.../f_intervals.json</code>
Table 7	nonlinear observables	replayed archive as above; empirical MSE by (4.2)
Fig. 5	bias-dominated regime	<code>round07_joint_cell/joint_cell.json</code>
Table 8	signed-structure map	<code>round21_applicability/applicability.json</code>
Prop. 3.5 check and share	is the identity right, and how much does it explain	<code>round25_finalstage/finalstage.json</code> ; aggregates from <code>round06_finalstep/finalstep.json</code>
Cor. 3.7, divergence fractions	when are the couplings identical	<code>round25_divergence/divergence.json</code> (200 seeds); <code>round22_corrections/corrections.json</code> (40 seeds)
Prop. 3.1 witness	does a replica match production	<code>round14_audit/audit.json</code> ; <code>single_vs_production</code> maximum absolute differences 0 in position, mass and field
Rem. 3.8	why there is no unconditional guarantee	<code>round07_counterexample/counterexample.json</code>
Table 9	are the intervals trustworthy	<code>round25_calibration/calibration.json</code> (supersedes the single-shape design in <code>round22_corrections/calibration.json</code>)